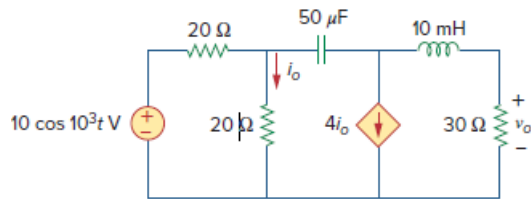


Problem

Use nodal analysis to find v_o in the circuit of Fig. 10.58.

Figure 10.58:



Step-by-step solution

Step 1 of 6

$$\omega = 10^3$$

$$10 \cos(10^3 t) \Rightarrow 10 \angle 0^\circ$$

$$10 \text{ mH} \Rightarrow j\omega L$$

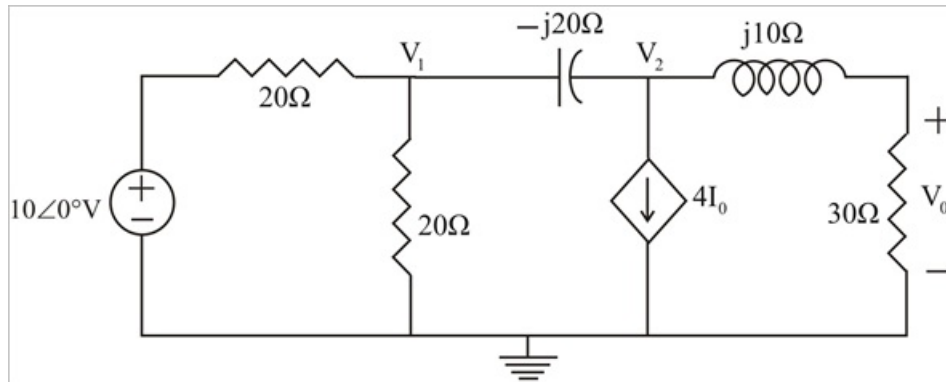
$$j\omega L = j10$$

$$50 \mu\text{F} \Rightarrow \frac{1}{j\omega C}$$

$$\frac{1}{j(10^3)(50 \times 10^{-6})} = -j20$$

Step 2 of 6

Consider the circuit shown below.



Step 3 of 6

At node 1,

$$\frac{10-V_1}{20} = \frac{V_1}{20} + \frac{V_1-V_2}{-j20}$$

$$\frac{10-V_1}{20} = \frac{V_1}{20} + \frac{j(V_1-V_2)}{20}$$

$$10-V_1 = V_1 + j(V_1-V_2)$$

$$10 = (2+j)V_1 - jV_2 \dots\dots\dots (1)$$

Step 4 of 6

At node 2,

$$\frac{V_1-V_2}{-j20} = 4I_0 + \frac{V_2}{30+j10}$$

Where,

$$I_0 = \frac{V_1}{20}$$

$$\frac{V_1-V_2}{-j20} = 4\left(\frac{V_1}{20}\right) + \frac{V_2}{30+j10}$$

$$\frac{j(V_1-V_2)}{20} = 4\left(\frac{V_1}{20}\right) + \frac{V_2}{30+j10}$$

$$V_1\left[\frac{j}{20} - \frac{4}{20}\right] = V_2\left[\frac{j}{20} + \frac{1}{30+j10}\right]$$

$$V_1(j-4) = V_2\left[\frac{j20}{20} + \frac{20}{30+j10}\right]$$

$$V_1(j-4) = V_2\left[j + \frac{20}{30+j10}\right]$$

$$V_1(j-4) = V_2\left[\frac{30j+j^210+20}{30+j10}\right]$$

$$V_1(j-4) = V_2\left[\frac{30j-10+20}{30+j10}\right] \quad (\because j^2 = -1)$$

$$V_1(j-4) = V_2\left[\frac{j30+10}{30+j10}\right]$$

$$(-4+j)V_1 = (0.6+j0.8)V_2$$

$$V_1 = \left(\frac{0.6+j0.8}{-4+j}\right)V_2 \dots\dots\dots (2)$$

Step 5 of 6

Substituting (2) into (1),

$$10 = (2 + j) \times \left[\frac{(0.6 + j0.8)}{-4 + j} \right] \times V_2 - jV_2$$

$$10 = \frac{(0.4 + j2.2)V_2}{-4 + j} - jV_2$$

$$10 = (0.035 - j0.541)V_2 - jV_2$$

$$10 = (0.035 - j0.541 - j)V_2$$

$$10 = (0.035 - j1.541)V_2$$

$$V_2 = \frac{10}{0.035 - j1.541}$$

Step 6 of 6

Therefore,

$$V_0 = \left(\frac{30}{30 + j10} \right) V_2$$

$$V_0 = \left(\frac{30}{30 + j10} \right) \left(\frac{10}{0.035 - j1.541} \right)$$

$$V_0 = \frac{300}{16.46 - j45.88}$$

$$V_0 = 2.0783 + j5.793$$

$$V_0 = 6.154 \angle 70.26^\circ \text{ V}$$

Therefore,

$$\therefore \boxed{V_0(t) = 6.154 \cos(10^3 t + 70.26^\circ) \text{ V}}$$